

Module 04 Lab - Exploratory Data

Analysis Objective: To learn how to explore a dataset to find patterns, anomalies, and insights before modeling. EDA is like being a detective; you are looking for clues in the data that will help you build a better model. **This lab is fully coded.** Your task is to run each cell, read the detailed explanations, understand the purpose of each visualization, and then complete the experimentation section.

Part 1: Setup and Data Loading**What is Exploratory Data Analysis (EDA)?**EDA is the process of using summary statistics and visualizations to understand a dataset's main characteristics. Before you can build a model, you need to understand your data. What stories does it tell? Are there errors or missing values? Are there strong relationships between variables? EDA helps answer these questions. We will use the famous Titanic dataset for this lab. It contains information about passengers and, crucially, whether they survived the disaster.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# Load the dataset directly from a URL
df = pd.read_csv('https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv')

print("---- First 5 Rows ----")
print(df.head())

print("\n--- Basic Info ---")
df.info()
```

---- First 5 Rows ----

	PassengerId	Survived	Pclass	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	

	Name	Sex	Age	SibSp
0	Braund, Mr. Owen Harris	male	22.0	1
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1
2	Heikkinen, Miss. Laina	female	26.0	0
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1
4	Allen, Mr. William Henry	male	35.0	0

	Parch	Ticket	Fare	Cabin	Embarked
0	0	A/5 21171	7.2500	NaN	S
1	0	PC 17599	71.2833	C85	C
2	0	STON/O2. 3101282	7.9250	NaN	S
3	0	113803	53.1000	C123	S
4	0	373450	8.0500	NaN	S

--- Basic Info ---

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 891 entries, 0 to 890
```

```
Data columns (total 12 columns):
```

#	Column	Non-Null Count	Dtype
0	PassengerId	891 non-null	int64
1	Survived	891 non-null	int64
2	Pclass	891 non-null	int64
3	Name	891 non-null	object
4	Sex	891 non-null	object
5	Age	714 non-null	float64
6	SibSp	891 non-null	int64
7	Parch	891 non-null	int64
8	Ticket	891 non-null	object
9	Fare	891 non-null	float64
10	Cabin	204 non-null	object
11	Embarked	889 non-null	object

```
dtypes: float64(2), int64(5), object(5)
```

```
memory usage: 83.7+ KB
```

Part 2: Descriptive Statistics Let's start by getting a high-level numerical summary of the data. The `.describe()`

- ✓ method is perfect for this. It calculates statistics like mean, standard deviation, min, and max for the numerical columns.

```
# Get summary statistics for numerical columns
print("---- Descriptive Statistics ----")
print(df.describe())

print("\n--- Key Insights from Statistics ---")
print(f"The average age of a passenger was {df['Age'].mean():.1f} years")
print(f"The overall survival rate was {df['Survived'].mean():.1%}.")
print(f"Fares ranged from ${df['Fare'].min()} to a whopping ${df['Fare'].max()}")
```

---- Descriptive Statistics ----

	PassengerId	Survived	Pclass	Age	SibSp	\
count	891.000000	891.000000	891.000000	714.000000	891.000000	
mean	446.000000	0.383838	2.308642	29.699118	0.523008	
std	257.353842	0.486592	0.836071	14.526497	1.102743	
min	1.000000	0.000000	1.000000	0.420000	0.000000	
25%	223.500000	0.000000	2.000000	20.125000	0.000000	
50%	446.000000	0.000000	3.000000	28.000000	0.000000	
75%	668.500000	1.000000	3.000000	38.000000	1.000000	
max	891.000000	1.000000	3.000000	80.000000	8.000000	

	Parch	Fare
count	891.000000	891.000000
mean	0.381594	32.204208
std	0.806057	49.693429
min	0.000000	0.000000
25%	0.000000	7.910400
50%	0.000000	14.454200
75%	0.000000	31.000000
max	6.000000	512.329200

--- Key Insights from Statistics ---

The average age of a passenger was 29.7 years.

The overall survival rate was 38.4%.

Fares ranged from \$0.0 to a whopping \$512.3292.

Part 3: Visual EDA - Telling Stories with Plots Numbers are great, but plots make patterns and relationships immediately obvious. The goal of visual EDA is to turn data

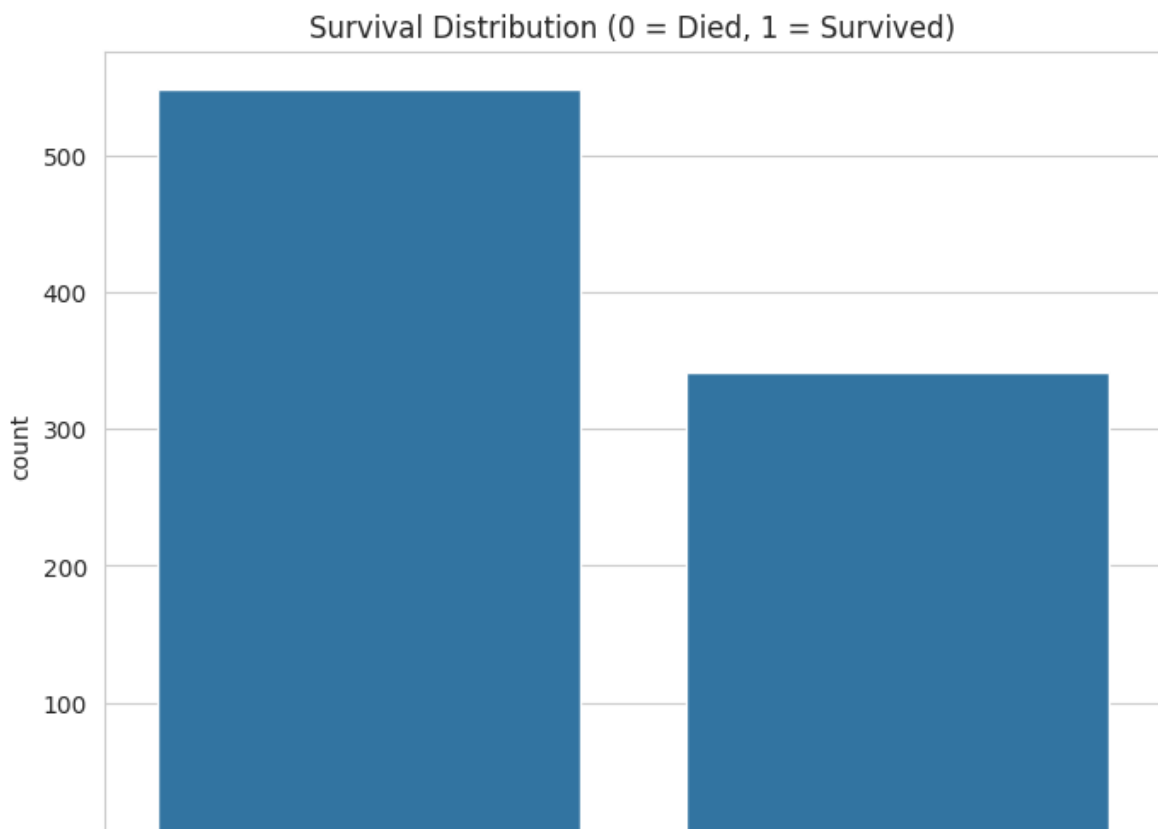
into insights. **A Note on Plotting Libraries:** * **Matplotlib:**

- ✓ The foundational library, gives you full control over everything. * **Seaborn:** Built on Matplotlib, it provides a simpler, high-level interface for creating common statistical plots. We will use Seaborn for its ease of use and attractive defaults.

Visualization 1: How many survived? A simple `countplot` is the best

- ✓ way to see the distribution of a categorical variable, like our target `Survived`.

```
sns.set_style('whitegrid')
plt.figure(figsize=(8, 6))
sns.countplot(x='Survived', data=df)
plt.title('Survival Distribution (0 = Died, 1 = Survived)')
plt.show()
print("Insight: Far more people died than survived. This is an example")
```

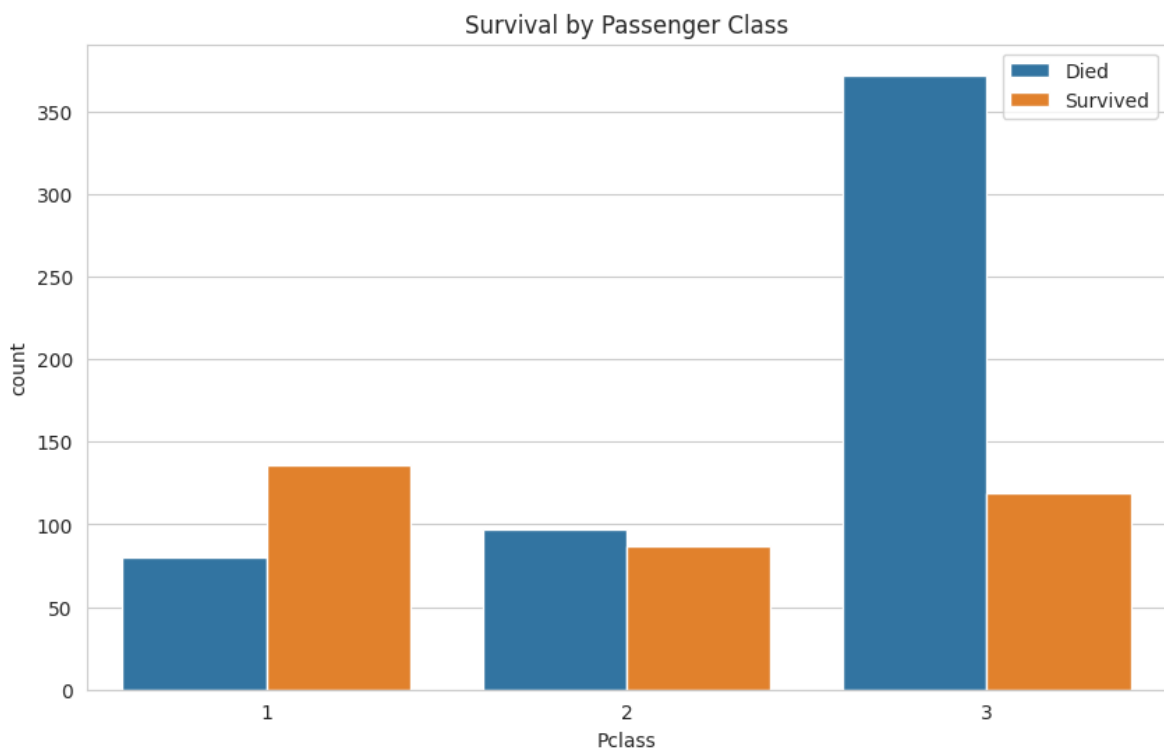




Insight: Far more people died than survived. This is an example of an

Visualization 2: Does passenger class matter for survival? Now we want to see if there's a relationship between two variables. We can use the `hue` parameter in `countplot` to split the bars by another category.

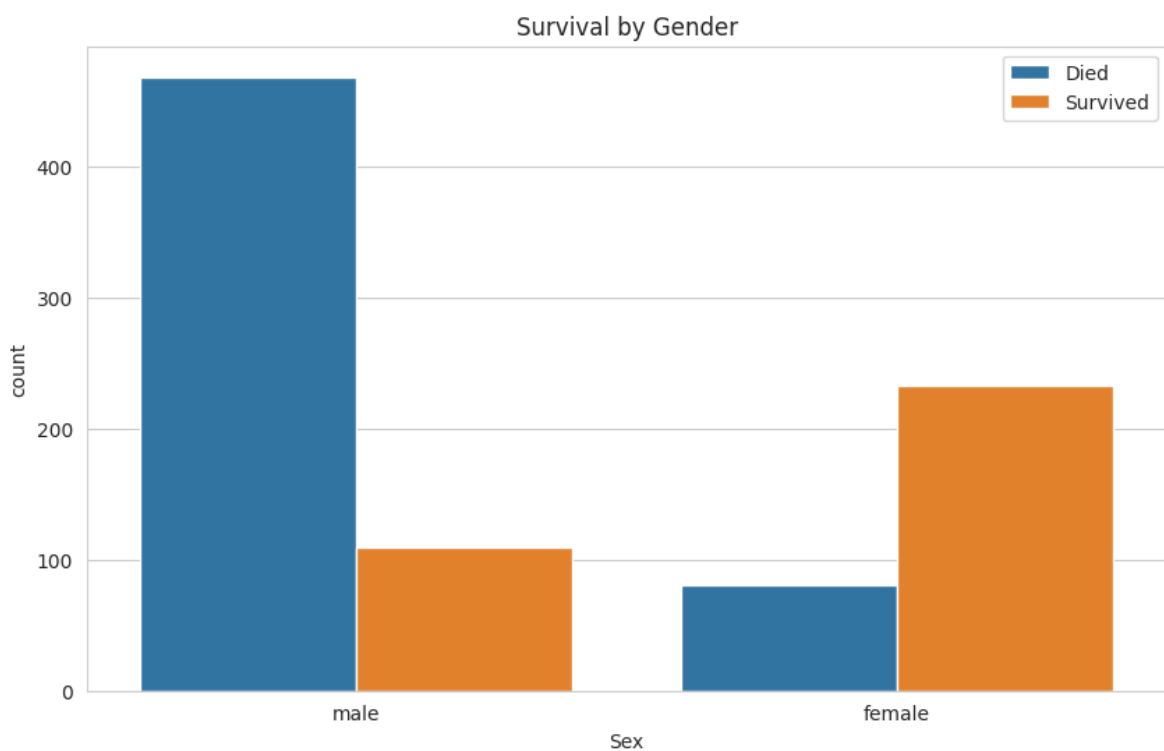
```
plt.figure(figsize=(10, 6))
sns.countplot(x='Pclass', hue='Survived', data=df)
plt.title('Survival by Passenger Class')
plt.legend(['Died', 'Survived'])
plt.show()
print("Insight: This is a very strong pattern. 1st class passengers h
```



Insight: This is a very strong pattern. 1st class passengers had a muc

- ✓ Visualization 3: What about gender? The 'women and children first' mantra is famous. Let's see if the data supports it.

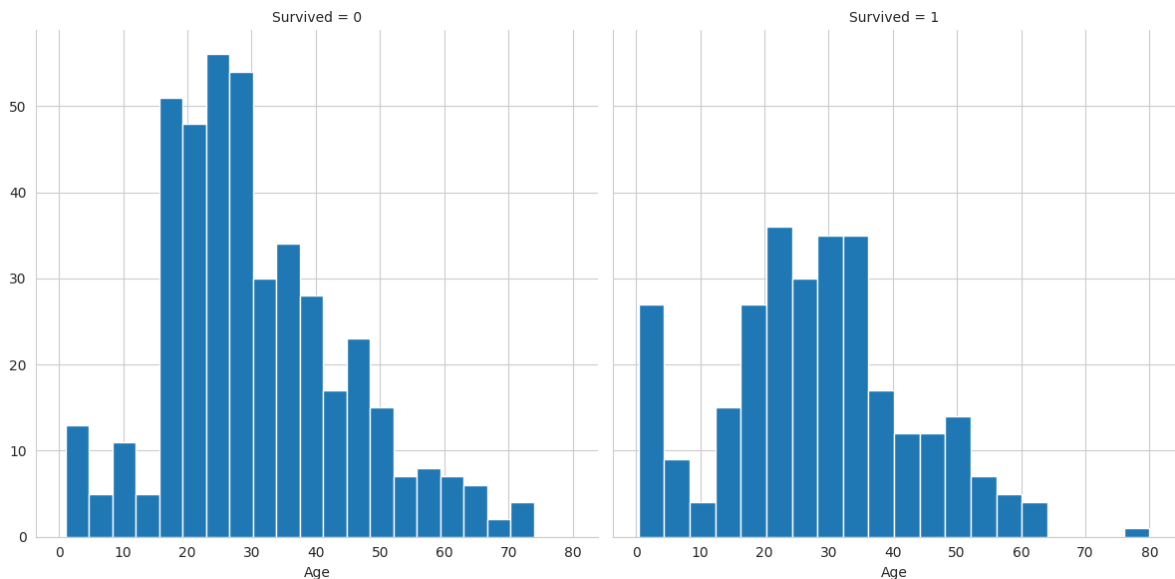
```
plt.figure(figsize=(10, 6))
sns.countplot(x='Sex', hue='Survived', data=df)
plt.title('Survival by Gender')
plt.legend(['Died', 'Survived'])
plt.show()
print("Insight: The pattern is undeniable. A much higher proportion of
```



Insight: The pattern is undeniable. A much higher proportion of female

Visualization 4: How does age play a role? For a continuous variable like `Age`, a histogram is a great way to see its distribution.

```
g = sns.FacetGrid(df, col='Survived', height=6)
g.map(plt.hist, 'Age', bins=20)
plt.show()
print("Insight: The age distribution for those who did not survive is
```



Insight: The age distribution for those who did not survive is centered

Part 4: Student Experimentation **Instructions:** Create your

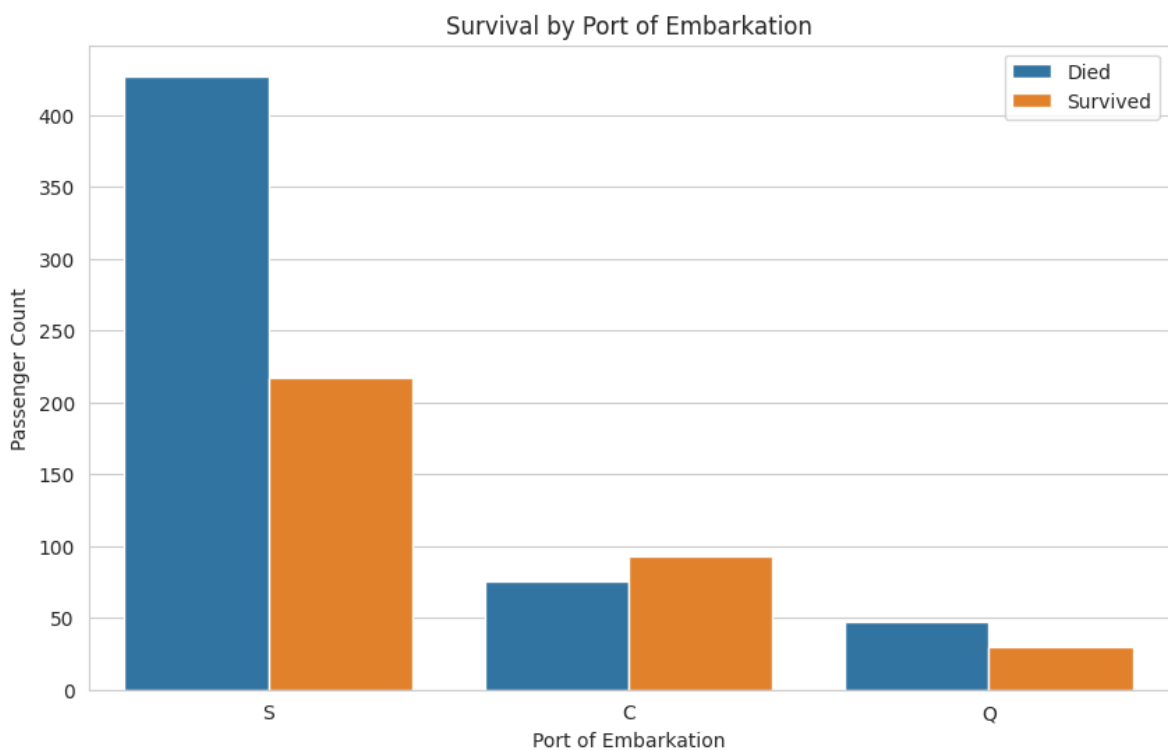
own visualizations to explore other relationships in the data.

Experiment 1: Port of Embarkation1. The `Embarked` column tells you where a passenger boarded the ship (C = Cherbourg, Q =

- ✓ Queenstown, S = Southampton).2. Create a `countplot` to see how survival rates differed by the port of embarkation. Does where they boarded seem to be related to their survival?

```
plt.figure(figsize=(10, 6))
sns.countplot(x='Embarked', hue='Survived', data=df)
plt.title('Survival by Port of Embarkation')
plt.xlabel('Port of Embarkation')
plt.ylabel('Passenger Count')
plt.legend(['Died', 'Survived'])
plt.show()
```

`print("Insight: Passengers who boarded at Cherbourg had a higher propo`

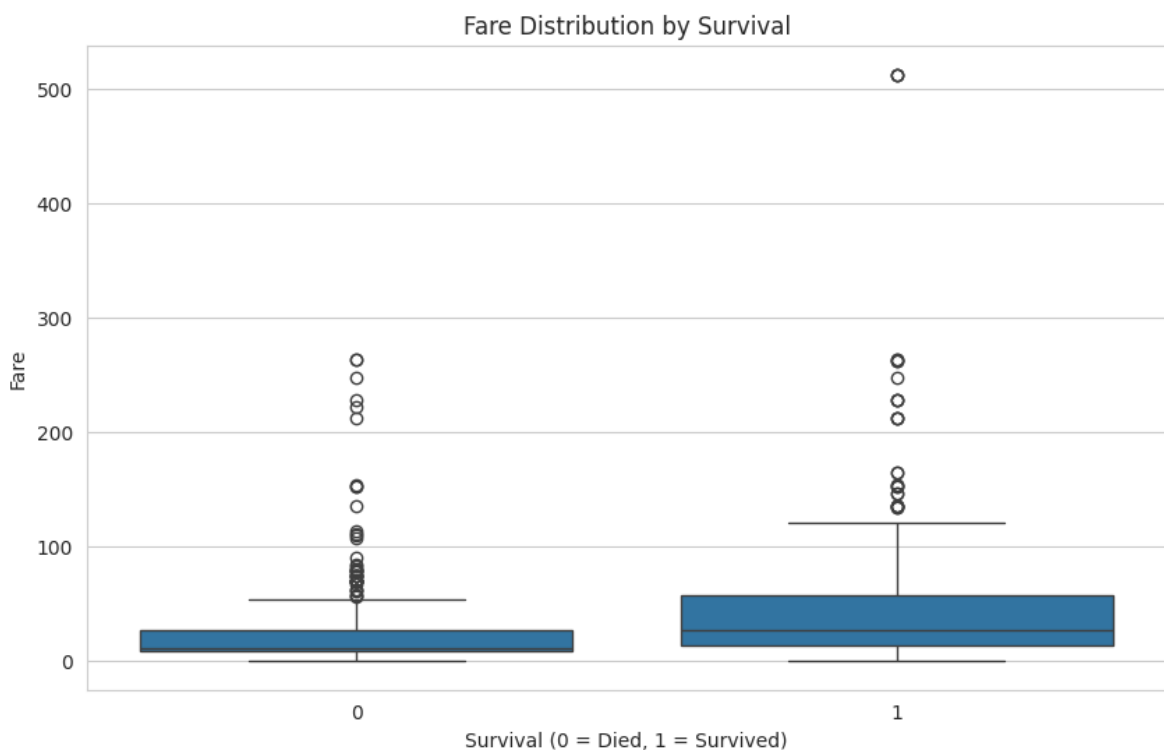


`Insight: Passengers who boarded at Cherbourg had a higher proportion o`

- Experiment 2: Fare vs. Survival1. **Fare** is a continuous numerical variable.2. A **boxplot** or **violinplot** is a great way to see the distribution of fares for those who survived vs. those who didn't.3. Create one of these plots to compare the **Fare** distribution by **Survived**. What does it tell you?

```
plt.figure(figsize=(10, 6))
sns.boxplot(x='Survived', y='Fare', data=df)
plt.title('Fare Distribution by Survival')
plt.xlabel('Survival (0 = Died, 1 = Survived)')
plt.ylabel('Fare')
plt.show()

print("Insight: Passengers who survived generally paid higher fares than those who died")
```



Insight: Passengers who survived generally paid higher fares than pass



Knowledge Check

Instructions: Answer the following questions in this markdown cell.

1. What is the primary goal of Exploratory Data Analysis (EDA)?

The primary goal of EDA is to understand a dataset before modeling by examining its structure, summary statistics, missing values, patterns, and relationships between variables.

2. Based on the plots in this lab, what kind of person had the best chance of surviving the Titanic?

Based on the plots, a female first-class passenger had the best chance of surviving. The age plots also showed a noticeable group of young children among the survivors.

3. Why is it important to visualize data instead of just looking at summary statistics? What can a plot show you that a number like 'mean' or 'count' can't?

Visualizations can reveal patterns, differences between groups, distributions, and outliers that a single summary number can hide. In this lab, the plots made the survival differences by class, gender, and age much easier to see than summary statistics alone.